

Activity

2/17/21

1. Medical image management

An X-ray image shows bones clearly, but the darker tissues have very poor visibility.

A) Which gray-level transformation would you recommend?

→ log transformation

B) Why is it suitable?

→ log transformation expands the intensity values in the dark regions while compressing the brighter regions.

→ This improves the visibility of soft tissues.

C) What effect will it have on the image?

Dark tissue become brighter, more detailed hidden structures in low-intensity regions become visible.

2. Night Time Surveillance:

A CCTV image has most pixel values b/w 0 and 40, making it very dark.

→ which transformation could improve the visibility
log transformation

B) why is it more suitable than negative transformation

→ log transformation enhances low-intensity pixels, making dark areas brighter

→ negative transformation only inverts brightness values and does not effectively enhance details in dark regions.

c) Predict the appearance of the Enhanced image

→ The parking lot will appear brighter

→ Vehicles, people and other objects become easier to identify

3. Finger Print Recognition.

→ A fingerprint image has fine ridge details, but the contrast b/w ridges and valleys is very low.

A) which gray-level transformation would you apply?

Histogram Equalization.

B) Explain why?

⇒ Histogram equalization redistributes the gray-level intensities across the entire intensity range.

⇒ It improves the global contrast of the fingerprint image.

C) what would happen if a log transformation were applied instead?

⇒ log transformation primarily enhances low intensity pixels.

⇒ It may brighten the fingerprint image but does not distribute the intensity values evenly.